Lab 1 Documentation

Domain Context

This mini-language is designed for data quality checks and corrections on integer datasets. It supports checking value ranges, membership in a list and duplicates, and allows invalid data to be corrected.

Statements

Assignment

Assignment stores a value or a calculation result in a variable or array element. It allows the program to correct invalid data, update values and save intermediate results for later use.

```
int quantity;
int correctedQuantity;

quantity = 12;
correctedQuantity = quantity + 3;
```

Input

The input statement reads an integer into a variable or array element. It allows the program to receive data for checking and correction and to process different values.

```
int quantity;
read(quantity);
```

Output

The output statement displays the value of an expression. It allows the user to see processed data, calculation results and the results of data quality checks.

```
int quantity;

read(quantity);
print(quantity);
```

Conditional Statement

The conditional statement chooses which instructions to execute based on a condition. It allows the program to handle valid and invalid data differently, applying corrections only when needed.

```
int quantity;

read(quantity);

if (quantity between (0, 100)) {
      print(quantity);
}
else {
      quantity = 0;
      print(quantity);
}
```

Loop

The loop repeats a sequence of statements while a condition is true. It allows the program to process array elements one by one and apply the same checks or corrections without repeating the same code.

```
int quantities[5];
int i;

i = 0;
while (i < 5) {
    read(quantities[i]);
    i = i + 1;
}
```